CRITICAL-PARAMETER RIGIDITY FOR THE RIEMANN ZETA FUNCTION UNDER L^2-PROJECTIVE COINCIDENCE

ROTKIV AYNÄSAC

Abstract. For a fixed interval between consecutive critical-line zeros of the Riemann zeta function, we use the Riemann–Siegel phase to transport every vertical fibre $\zeta(\beta + it)$, $1/2 \le \beta \le 1$, to one common parameter interval. In the Hilbert space $L^2([0,1])$ we remove the four-dimensional polynomial subspace of degree at most three and then projectivize. The main unconditional result is a rigidity theorem: two projected vertical zeta fibres have the same projective class if and only if their real parameters are equal. In particular, coincidence with the class at $\beta = 1/2$ forces $\beta = 1/2$. The proof is global and elementary once the phase coordinate has been fixed: a hypothetical projective coincidence produces a finite-dimensional functional identity, analytic continuation transports it to the simple pole of ζ at $s = 1$, and the pole cannot be cancelled by the remaining terms.

We then separate this genuine zeta-specific rigidity from the universal holomorphic transport equation. The entire function $g(s) = s - s_0$ satisfies exactly the same Cauchy–Riemann transport law, conformality, positive Jacobian and harmonicity, while possessing a zero at any prescribed $\beta_0 \neq 1/2$. Thus these universal geometric identities cannot by themselves confine zeros to the critical line. The missing zero-specific step is isolated as an exact projective-collapse condition. We prove that this condition is equivalent to the Riemann hypothesis, and also give finite-dimensional, fourth-derivative and Wronskian formulations of the same condition. Finally, a quantitative version based on the Fubini–Study defect shows how a zero-triggered *almost* coincidence, if established by a zeta-specific arithmetic estimate, would force quantitative proximity to the critical line without assuming exact collapse.

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1. Introduction

Write

$$s = \beta + it, \qquad \frac{1}{2} \le \beta \le 1, \tag{1.1}$$

and consider the vertical family, in the setting initiated by Riemann's original memoir [33],

$$\boxed{\zeta(\beta + it).} \tag{1.2}$$

The object studied in this paper is the Riemann zeta function itself. No auxiliary replacement zeta function is introduced. On each interval between consecutive critical-line zeros, the Riemann–Siegel phase supplies a common coordinate; the corresponding vertical restrictions are then placed in the Hilbert space $L^2([0,1])$, quotiented by the four-dimensional polynomial space of degree at most three, and finally projectivized. All three operations are fixed before the real parameter β is varied.

The central unconditional result is the global rigidity statement

$$\boxed{[Q_\perp \zeta(\alpha + it_n(\cdot))] = [Q_\perp \zeta(\beta + it_n(\cdot))] \quad \Longleftrightarrow \quad \alpha = \beta,} \tag{1.3}$$

valid on every fixed phase cell. In particular,

$$\boxed{[Q_\perp \zeta(\beta + it_n(\cdot))] = [Q_\perp \zeta(\tfrac{1}{2} + it_n(\cdot))] \quad \Longleftrightarrow \quad \beta = \frac{1}{2}.} \tag{1.4}$$

The proof is exact rather than asymptotic. Projective coincidence produces a finite-dimensional identity; analytic continuation carries that identity to the simple pole of ζ at $s = 1$; and the pole cannot be cancelled by the remaining fibre or by the discarded cubic component.

1.1. Where this construction sits among classical RH criteria.

The present construction lies at the intersection of two old and substantial traditions, but is not a restatement of either one. The first is the Hilbert-space tradition surrounding the Nyman–Beurling criterion and its arithmetic refinements: RH can be encoded as an L^2 closure problem for fractional-part dilates [30, 7, 3, 4, 5, 6]. The second is the complex-analytic rigidity tradition going back to Ostrowski and Voronin and continuing through modern differential, functional, and difference-independence results for ζ [31, 38, 22, 9, 28, 23]. Here the Hilbert-space quotient is local in a Riemann–Siegel phase cell, whereas the decisive uniqueness mechanism is meromorphic: the pole at $s = 1$ separates every two distinct vertical parameters after the quotient.

A closely related older line is the geometric/derivative route represented by Speiser's criterion and the quantitative work of Levinson–Montgomery on zeros of derivatives of ζ [34, 26]. Those results translate information about the zero set into restrictions on ζ' and its derivatives. The present construction does something different but complementary: differentiation is taken only after a fixed Riemann–Siegel phase reparametrization, the polynomial components of degree at most three are removed in L^2, and uniqueness is then tested projectively. Thus the paper combines two classical ideas—Hilbert-space reduction and analytic zero/derivative rigidity—without reducing to either classical criterion.

This should also be distinguished from the global positivity/equivalent- criterion tradition of Weil and Li [39, 27, 8]. Those criteria encode all zeros through global test functions, explicit formulae,

or positivity sequences. The present criterion instead compares two vertical zeta fibres on one fixed cell and asks whether their classes in a finite-codimensional projective Hilbert quotient can coincide. Within the literature reviewed here, we have not found a criterion with the simultaneous combination

$$\boxed{\text{Riemann–Siegel cell} + L^2/\mathcal{P}_3 + \text{projective coincidence} + \text{pole separation at } s = 1.} \tag{1.5}$$

The components are classical; the way they are coupled in (1.5), and the resulting exact critical-parameter rigidity, are the distinctive content of the construction.

The unconditional theorem has value independently of the unresolved RH trigger. It identifies a concrete one-parameter family in an infinite- dimensional Hilbert space for which quotienting by a fixed four-dimensional subspace and then forgetting scalar magnitude still loses no information about the horizontal parameter. That injectivity is not a generic Hilbert- space fact: it is enforced here by the global meromorphic singularity of ζ. In particular, the construction yields an exact bridge from local L^2 projective geometry to global analytic continuation, rather than an asymptotic numerical separation. The RH-equivalent condition introduced later should therefore be read as an additional arithmetic trigger attached to an already nontrivial rigidity theorem, not as the source of the theorem's content.

There is a second neighboring literature that must be kept logically separate. Values of ζ or its derivatives at shifted zero ordinates have been studied through discrete mean values, universality, and local value distribution [19, 15, 16, 24, 10, 25, 17, 18]. These results contain rich information about $\zeta(\beta+i\gamma)$ and nearby shifts, but none of the cited theorems turns the scalar event $\zeta(\beta+i\gamma) = 0$ into the exact whole-cell finite-dimensional relation used below. Difference-independence, on the other hand, supports the rigidity side of the paper, not the missing zero-trigger.

Finite-dimensional translation-invariant spaces supply another relevant background: such spaces are governed by exponential-polynomial structure [2, 13, 35]. This makes a finite-rank collapse a natural object to test, but it does not make a point zero force that collapse. The distinction between *classification of a finite-dimensional relation* and *generation of that relation by a zero* is one of the main logical separations of this paper.

1.2. **Why the route is not a zero-free-region approach to** $1/2$**.** A classical route to RH proceeds through prime-number-theorem error terms or zero-free regions. Von Koch connected RH with square-root-scale prime counting errors [37]; modern unconditional zero-free regions remain of de la Vallée Poussin or Vinogradov–Korobov type and approach $\Re s = 1$ rather than $\Re s = 1/2$ [14, 29]. In that framework, improving an error exponent all the way to the critical exponent $1/2$ is not an innocuous limiting process: at the relevant uniform strength it is itself an RH-level statement. A hypothetical zero with real part $\beta > 1/2$ produces an x^β contribution in the explicit formula and cannot remain uniformly hidden below a square-root-scale error.

The present route does not attempt to move a contour through a sequence of zero-free strips, nor does it require a bound of the form $\beta \le 1/2 + \delta(T)$ with $\delta(T)$ shrinking logarithmically. Its unconditional separation is exact for every fixed $\beta > 1/2$:

$$M_n(\beta) > 0 \qquad (\beta > 1/2), \tag{1.6}$$

while at the same time

$$M_n(\beta) = P_n(1/2)(\beta - 1/2)^2 + o\Big((\beta - 1/2)^2\Big) \qquad (\beta \downarrow 1/2). \tag{1.7}$$

Thus the projective classes may approach the critical class arbitrarily closely, and in fact do so quadratically, but they cannot coincide at any $\beta > 1/2$. No uniform positive gap from $1/2$ is claimed or needed. This is precisely why the usual "how close can one push a zero-free boundary to $1/2$?" obstruction does not condemn the rigidity theorem: the theorem is a nonperturbative uniqueness statement, not a zero-free-region estimate.

This does not remove the difficulty of RH; it relocates it. The projective side is closed, but the missing implication is

$$\zeta(\beta+i\gamma)=0 \quad\Longrightarrow\quad [Q_\perp\zeta(\beta+it_n(\cdot))]=[Q_\perp\zeta(\tfrac12+it_n(\cdot))]. \tag{1.8}$$

Theorem 11.1 proves that (1.8) is equivalent to RH once the unconditional rigidity is available. It is therefore not presented as a small technical lemma. What the construction achieves is a sharp localization of the RH content: everything to the right of the zero-trigger is rigid and explicit; what remains is to explain why the arithmetic event "ζ has a zero" should force exact or quantitative projective collapse.

1.3. **A possible intermediate target: almost-coincidence.** The exact implication (1.8) is RH-equivalent, but the geometry also produces a weaker target that is not tautologically RH. If one can prove, at a zero,

$$M_n(\beta)\le \varepsilon_n(\gamma), \tag{1.9}$$

then the already proved lower rigidity converts this into a bound for $|\beta-1/2|$. Locally,

$$\left|\beta-\frac12\right|\le\sqrt{\frac{2\varepsilon_n(\gamma)}{P_n(1/2)}} \tag{1.10}$$

under the explicit threshold stated in Theorem 12.2. This gives the construction independent value even before exact collapse: a zeta-specific arithmetic upper bound for the projective defect would immediately become a quantitative horizontal bound for zeros. It is a different object from a smooth fit to zero ordinates and from a classical zero-free strip.

Finally, the universal Cauchy–Riemann transport must not be confused with the zeta-specific rigidity. Section 8 gives the entire counterexample $g(s)=s-s_0$: it obeys the same transport equation, conformality, positive Jacobian, and harmonicity, while having a zero at any prescribed $\beta_0\neq 1/2$. Hence no proof of (1.8) can be built from those universal geometric identities alone. The role of the pole at $s=1$ is therefore twofold: it proves the exact projective uniqueness, and it marks the first genuinely zeta-specific ingredient in the route.

2. Analytic setup and the common phase coordinate

We use standard facts about ζ, Hardy's real Z-function, the Riemann–Siegel phase and the functional equation as in [36, 21, 12]; special-function identities for Γ, ψ and their derivatives are also recorded in the NIST DLMF [11]. Define

$$\vartheta(t)=\Im\log\Gamma\left(\frac14+\frac{it}{2}\right)-\frac{t}{2}\log\pi, \tag{2.1}$$

with the standard continuous branch on the positive real axis. Then

$$\zeta\left(\frac12+it\right)=Z(t)e^{-i\vartheta(t)},\qquad Z(t)\in\mathbb{R}. \tag{2.2}$$

Let

$$0<\gamma_1<\gamma_2<\cdots \tag{2.3}$$

be the increasing sequence of distinct positive ordinates for which $\zeta(1/2+i\gamma_n)=0$. Hardy's theorem gives infinitely many such ordinates [20]; discreteness of zeros implies that they have no finite accumulation point [36].

Lemma 2.1 (Positivity of the phase derivative on every positive zero cell)**.** *For every $n\ge1$,*

$$\vartheta'(t)>0\qquad(\gamma_n\le t\le\gamma_{n+1}). \tag{2.4}$$

Consequently

$$\Omega_n:=\vartheta(\gamma_{n+1})-\vartheta(\gamma_n)>0. \tag{2.5}$$

Proof. Differentiating (2.1) gives

$$\vartheta'(t) = \frac{1}{2}\Re\psi\Big(\frac{1}{4}+\frac{it}{2}\Big) - \frac{1}{2}\log\pi, \tag{2.6}$$

$$\vartheta''(t) = -\frac{1}{4}\Im\psi'\Big(\frac{1}{4}+\frac{it}{2}\Big). \tag{2.7}$$

For $\Re z > 0$ the trigamma series

$$\psi'(z) = \sum_{k=0}^{\infty} \frac{1}{(k+z)^2} \tag{2.8}$$

is absolutely convergent. With $z = 1/4 + it/2$ and $t > 0$,

$$\Im\frac{1}{(k+\frac{1}{4}+\frac{it}{2})^2} = -\frac{(k+\frac{1}{4})t}{((k+\frac{1}{4})^2+t^2/4)^2} < 0. \tag{2.9}$$

Absolute convergence permits termwise summation, so $\vartheta''(t) > 0$ for $t > 0$. Hence ϑ' is strictly increasing.

A rigorous bound follows by truncating

$$\psi(z) = -\gamma_E + \sum_{k=0}^{\infty}\Big(\frac{1}{k+1} - \frac{1}{k+z}\Big) \tag{2.10}$$

at $k = 199$. With $z = \frac{1}{4} + 7i$ (i.e. $t = 14$), write

$$\Re\frac{1}{k+z} = \frac{k+\frac{1}{4}}{(k+\frac{1}{4})^2+49}, \tag{2.11}$$

so that the partial sum

$$S_{199} := \sum_{k=0}^{199}\left(\frac{1}{k+1} - \frac{k+\frac{1}{4}}{(k+\frac{1}{4})^2+49}\right) = 2.526047\ldots \tag{2.12}$$

is an explicit finite rational-arithmetic quantity. For the tail, $|\Re[\frac{1}{k+1} - \frac{1}{k+z}]| \le \frac{3}{4}/(k+\frac{1}{4})$ for $k \ge 200$ (both terms are $O(1/k)$ with the stated constant once $k \ge 200 > 49$), giving

$$\left|\sum_{k=200}^{\infty}\Re\Big(\frac{1}{k+1}-\frac{1}{k+z}\Big)\right| \le \frac{3}{4}\sum_{k=200}^{\infty}\frac{1}{(k+\frac{1}{4})^2} \le \frac{3}{4\cdot 199} = 0.0037688\ldots. \tag{2.13}$$

Combining (2.12) and (2.13) with $\gamma_E = 0.5772156\ldots$ gives the certified interval

$$\Re\psi\big(\tfrac{1}{4}+7i\big) \in [1.945063,\ 1.952600], \tag{2.14}$$

and hence, from (2.6) with $\log\pi = 1.144730\ldots$,

$$\vartheta'(14) \in [0.400166,\ 0.403935] \subset (0.40016, \infty). \tag{2.15}$$

This certified interval contains the independently computed reference value $\vartheta'(14) = 0.4004837\ldots$, confirming the bound. The first positive critical-line zero satisfies $\gamma_1 = 14.134725\ldots > 14$; this is also contained in rigorous computational verifications of the critical line, far beyond this height [32]. Since ϑ' is increasing, (2.4) follows for every $t \ge \gamma_1$, and then (2.5) follows by integration. □

Definition 2.2 (Common phase coordinate)**.** *For a fixed n, define $t_n : [0,1] \to [\gamma_n, \gamma_{n+1}]$ by*

$$\boxed{\vartheta(t_n(q)) = \vartheta(\gamma_n) + \Omega_n q.} \tag{2.16}$$

Then t_n is a real-analytic increasing diffeomorphism and

$$\boxed{t_n'(q) = \frac{\Omega_n}{\vartheta'(t_n(q))} > 0.} \tag{2.17}$$

For $1/2 \le \beta \le 1$ we write, only as a shorthand for the same zeta function,

$$\zeta_{\beta,n}(q) := \zeta(\beta + it_n(q)). \tag{2.18}$$

Proof. Lemma 2.1 and the inverse function theorem give a unique real-analytic inverse of ϑ on the compact interval. Differentiating (2.16) gives (2.17). Since $t_n([0,1]) \subset (0,\infty)$, none of the points $\beta + it_n(q)$ equals the pole $s = 1$, so (2.18) is real-analytic in q and smooth jointly in (β, q) on $[1/2, 1] \times [0, 1]$. □

3. Universal holomorphic transport and its geometric consequences

The following identities are universal. They do not use the Euler product, functional equation, zero distribution, or pole structure of ζ.

Theorem 3.1 (Universal quarter-turn geometry)**.** *Let g be holomorphic on a domain containing a rectangle in the variable $s = \beta + it$, and define $G(\beta, t) = g(\beta + it)$. Then*

$$\boxed{G_\beta = -iG_t.} \tag{3.1}$$

At every point,

$$|G_\beta| = |G_t|, \tag{3.2}$$

$$\Re(\overline{G_\beta} G_t) = 0, \tag{3.3}$$

$$J_G = |G_t|^2 = |g'(\beta + it)|^2 \ge 0, \tag{3.4}$$

$$G_{\beta\beta} + G_{tt} = 0. \tag{3.5}$$

Where $g' \neq 0$, the pullback of the Euclidean metric is

$$\boxed{g^*(|dw|^2) = |g'(\beta + it)|^2 (\, d\beta^2 + \, dt^2).} \tag{3.6}$$

Proof. Holomorphy gives $G_\beta = g'(\beta + it)$ and $G_t = ig'(\beta + it)$, proving (3.1). The norm and orthogonality identities follow immediately. The real Jacobian of a holomorphic map is $|g'|^2$, and (3.5) follows by differentiating (3.1). The metric identity follows by evaluating the squared norm of $g'(\beta + it)(\, d\beta + i\, dt)$. □

Corollary 3.2 (Exact transport in the fixed phase coordinate)**.** *For any such g, put*

$$G_n(\beta, q) := g(\beta + it_n(q)). \tag{3.7}$$

Then

$$\boxed{\partial_\beta G_n(\beta, q) = -ia_n(q)\partial_q G_n(\beta, q), \qquad a_n(q) := \frac{1}{t_n'(q)} = \frac{\vartheta'(t_n(q))}{\Omega_n}.} \tag{3.8}$$

Proof. The chain rule gives $\partial_q G_n = it_n'(q) g'(\beta + it_n(q))$ and $\partial_\beta G_n = g'(\beta + it_n(q))$. Since $t_n'(q) > 0$, division by $t_n'(q)$ is legitimate and yields (3.8). □

4. The L2 quotient and projective defect

Let

$$\mathcal{H} = L^2([0,1],\, dq; \mathbb{C}), \qquad \langle f, h\rangle = \int_0^1 \overline{f(q)} h(q)\, dq, \tag{4.1}$$

with inner product linear in the second variable. Let

$$\mathcal{P}_3 = \operatorname{span}\{1, q, q^2, q^3\}, \qquad Q_\perp = I - \Pi_3, \tag{4.2}$$

where Π_3 is the orthogonal projection onto $\mathcal{P}_3$.

Lemma 4.1 (Quotient interpretation)**.** *The range of $Q_\perp$ is $\mathcal{P}_3^\perp$ and its kernel is $\mathcal{P}_3$. For $f, h \in \mathcal{H}$,*

$$\langle [f], [h] \rangle_{\mathcal{H}/\mathcal{P}_3} := \langle Q_\perp f, Q_\perp h \rangle \tag{4.3}$$

defines the canonical Hilbert-space structure on $\mathcal{H}/\mathcal{P}_3$.

Proof. This is the orthogonal decomposition $\mathcal{H} = \mathcal{P}_3 \oplus \mathcal{P}_3^\perp$. Each quotient class has exactly one representative in $\mathcal{P}_3^\perp$, namely $Q_\perp f$, and (4.3) is independent of the chosen representative. □

For a nonzero $u \in \mathcal{H}$, let $[u] \in \mathbb{P}(\mathcal{H})$ denote its complex projective class. Once nonvanishing has been proved, define

$$u_{\beta,n} := Q_\perp \zeta_{\beta,n} \tag{4.4}$$

and the Fubini–Study defect from the critical fibre by

$$\boxed{M_n(\beta) = 1 - \frac{|\left\langle u_{1/2,n}, u_{\beta,n} \right\rangle|^2}{\left\|u_{1/2,n}\right\|^2 \|u_{\beta,n}\|^2}.} \tag{4.5}$$

Thus $0 \le M_n(\beta) \le 1$, and $M_n(\beta) = 0$ exactly when the two nonzero vectors have the same projective class.

5. Holomorphic continuation of the phase coordinate

The pole-separation argument requires a complex neighborhood of the real phase cell. Define

$$\Theta(z) = \frac{1}{2i}\left[\log\Gamma\left(\frac{1}{4} + \frac{iz}{2}\right) - \log\Gamma\left(\frac{1}{4} - \frac{iz}{2}\right)\right] - \frac{z}{2}\log\pi \tag{5.1}$$

on

$$S := \{z \in \mathbb{C} : |\Im z| < 1/2\}. \tag{5.2}$$

Both Gamma arguments have positive real part on S. Since Γ has no zeros and no poles in the open right half-plane, holomorphic logarithms may be chosen there; these choices can be normalized so that $\Theta(t) = \vartheta(t)$ for real t.

Define

$$q_n(z) = \frac{\Theta(z) - \Theta(\gamma_n)}{\Omega_n}. \tag{5.3}$$

Then q_n is holomorphic on S and equals the real phase coordinate on the cell.

Lemma 5.1 (Boundary growth at the critical pole)**.** *Let $z_0 = -i/2$. As $z \to z_0$ from within S,*

$$q_n(z) = O(1 + |\log(z - z_0)|)\,. \tag{5.4}$$

Consequently, for every $p_3 \in \mathcal{P}_3$,

$$p_3(q_n(z)) = O\Big(1 + |\log(z - z_0)|^3\Big)\,. \tag{5.5}$$

Proof. At $z = z_0$, the factor $1/4 - iz/2$ tends to 0, whereas $1/4 + iz/2$ tends to $1/2$. In the right half-plane, the Laurent expansion $\Gamma(w) = w^{-1} + O(1)$ implies $\log\Gamma(w) = -\log w + O(w)$ on a fixed local branch. Therefore (5.1) has at most logarithmic growth at z_0, which gives (5.4). A polynomial of degree at most three then satisfies (5.5). □

6. Pole separation and L2-projective rigidity

Lemma 6.1 (Pole separation)**.** *Let*

$$\frac{1}{2} \le \alpha < \beta \le 1, \qquad \lambda \in \mathbb{C}^\times, \qquad p_3 \in \mathcal{P}_3. \tag{6.1}$$

Then the identity

$$\zeta(\alpha + it_n(q)) - \lambda\zeta(\beta + it_n(q)) = p_3(q) \tag{6.2}$$

cannot hold on any nonempty real subinterval of $[0, 1]$.

Proof. If (6.2) holds in L^2 on an interval, the difference of the two sides is continuous and vanishes almost everywhere. If it were nonzero at one point, continuity would make it nonzero on an open subinterval of positive measure, a contradiction. Thus the identity is pointwise on the interval.

Replacing q by $q_n(z)$ on the corresponding real z-interval gives

$$\zeta(\alpha + iz) - \lambda\zeta(\beta + iz) = p_3(q_n(z)). \tag{6.3}$$

Both sides are meromorphic on the connected strip S. The identity theorem for meromorphic functions [1] therefore extends (6.3) throughout S.

Set

$$z_\beta = -i(1 - \beta). \tag{6.4}$$

Because $\beta > \alpha \ge 1/2$, one has $\beta > 1/2$ and hence $z_\beta \in S$. Moreover

$$\beta + iz_\beta = 1, \qquad \alpha + iz_\beta = 1 - (\beta - \alpha) \ne 1. \tag{6.5}$$

Thus the second zeta term has a simple pole at z_β, while the first zeta term and $p_3(q_n(z))$ are holomorphic there. Since $\lambda \ne 0$, the pole cannot cancel. This contradiction proves the lemma. □

Proposition 6.2 (Nonvanishing of every projected zeta fibre)**.** *For every* $n \ge 1$ *and every* $1/2 \le \beta \le 1$*,*

$$\boxed{Q_\perp\zeta(\beta + it_n(\cdot)) \ne 0 \quad \text{in } L^2([0, 1]).} \tag{6.6}$$

Proof. Suppose first that $\beta > 1/2$ and $Q_\perp\zeta_{\beta,n} = 0$. Then $\zeta_{\beta,n} = p_3$ in L^2 for some $p_3 \in \mathcal{P}_3$. Continuity makes the identity pointwise. Analytic continuation gives

$$\zeta(\beta + iz) = p_3(q_n(z)) \tag{6.7}$$

on S. At $z = -i(1 - \beta) \in S$ the left side has the simple pole of ζ at 1, while the right side is holomorphic, a contradiction.

Now let $\beta = 1/2$. The same assumption and identity theorem give

$$\zeta\left(\frac{1}{2} + iz\right) = p_3(q_n(z)) \tag{6.8}$$

on S. Write $z = z_0 + w$ with $z_0 = -i/2$. Then

$$\zeta\left(\frac{1}{2} + i(z_0 + w)\right) = \zeta(1 + iw) = \frac{1}{iw} + O(1), \tag{6.9}$$

whereas Lemma 5.1 gives only logarithmic-cubic growth on the right side of (6.8). Since $|w|^{-1}/(1 + |\log|w||^3) \to \infty$, the identity is impossible. □

Theorem 6.3 (Global L^2-projective rigidity)**.** *For every fixed phase cell and all* $\alpha, \beta \in [1/2, 1]$*,*

$$\boxed{[Q_\perp\zeta(\alpha + it_n(\cdot))] = [Q_\perp\zeta(\beta + it_n(\cdot))] \quad \Longleftrightarrow \quad \alpha = \beta.} \tag{6.10}$$

Proof. The reverse implication is immediate. Suppose the projective classes are equal and $\alpha \neq \beta$. By interchanging the two parameters, assume $\alpha < \beta$. Proposition 6.2 ensures that both classes are defined. Equality in projective space gives $\lambda \in \mathbb{C}^\times$ such that

$$Q_\perp \zeta_{\alpha,n} = \lambda Q_\perp \zeta_{\beta,n}. \tag{6.11}$$

Hence

$$Q_\perp(\zeta_{\alpha,n} - \lambda \zeta_{\beta,n}) = 0. \tag{6.12}$$

Since $\ker Q_\perp = \mathcal{P}_3$, there is $p_3 \in \mathcal{P}_3$ such that (6.2) holds. This contradicts Lemma 6.1. □

Corollary 6.4 (Critical-parameter rigidity)**.** *For every fixed phase cell and every* $1/2 \leq \beta \leq 1$,

$$\boxed{[Q_\perp \zeta(\beta + it_n(\cdot))] = [Q_\perp \zeta(\tfrac{1}{2} + it_n(\cdot))] \quad \Longleftrightarrow \quad \beta = \frac{1}{2}.} \tag{6.13}$$

Equivalently,

$$\boxed{M_n(\beta) = 0 \quad \Longleftrightarrow \quad \beta = \frac{1}{2}.} \tag{6.14}$$

Proof. Take $\alpha = 1/2$ in Theorem 6.3. The equivalence with (6.14) follows from equality in the Cauchy–Schwarz inequality for two nonzero vectors. □

Proposition 6.5 (Global projective immersion)**.** *For every fixed phase cell and every* $1/2 \leq \beta \leq 1$,

$$u_{\beta,n} \wedge \partial_\beta u_{\beta,n} \neq 0. \tag{6.15}$$

Equivalently,

$$P_n(\beta) := \frac{\|u_{\beta,n}\|^2 \, \|\partial_\beta u_{\beta,n}\|^2 - |\langle u_{\beta,n}, \partial_\beta u_{\beta,n}\rangle|^2}{\|u_{\beta,n}\|^4} > 0. \tag{6.16}$$

Proof. The map $\beta \mapsto u_{\beta,n}$ is C^∞ as an L^2-valued map, because all joint derivatives of $\zeta(\beta + it_n(q))$ are continuous on the compact parameter rectangle. Suppose (6.15) fails. Since $u_{\beta,n} \neq 0$, there is $c \in \mathbb{C}$ with

$$Q_\perp\big(\zeta'(\beta + it_n(\cdot)) - c\zeta(\beta + it_n(\cdot))\big) = 0. \tag{6.17}$$

Thus for some $p_3 \in \mathcal{P}_3$,

$$\zeta'(\beta + it_n(q)) - c\zeta(\beta + it_n(q)) = p_3(q). \tag{6.18}$$

For $\beta > 1/2$, analytic continuation to $z = -i(1-\beta)$ produces a double pole from ζ' and at most a simple pole from $c\zeta$, while the right side is holomorphic. The double pole cannot cancel.

For $\beta = 1/2$, continue to S and let $z = z_0 + w$. The Laurent expansion at $s = 1$ gives

$$\zeta'(1 + iw) = \frac{1}{w^2} + O(1), \qquad \zeta(1 + iw) = \frac{1}{iw} + O(1), \tag{6.19}$$

whereas the right side is $O(1 + |\log w|^3)$. Again the leading pole cannot cancel. This proves (6.15); the strict positivity of (6.16) is the strict Cauchy–Schwarz inequality. □

7. Recoverability and quantitative separation

Theorem 7.1 (Exact recoverability on the zeta projective image)**.** *For fixed* n*, define*

$$\Phi_n : [1/2, 1] \longrightarrow \mathbb{P}(\mathcal{H}), \qquad \Phi_n(\beta) = [u_{\beta,n}]. \tag{7.1}$$

Then Φ_n *is a continuous injective immersion and a homeomorphism onto its image. Hence a projective class in* $\Phi_n([1/2,1])$ *uniquely determines* β*, and hence the complete unprojected restriction* $q \mapsto \zeta(\beta + it_n(q))$*.*

Proof. Continuity follows from continuity of $\beta \mapsto u_{\beta,n}$ and the fact that it never vanishes. Injectivity is Theorem 6.3, and immersion is Proposition 6.5. The interval $[1/2, 1]$ is compact and projective Hilbert space equipped with the Fubini–Study metric is Hausdorff. Therefore a continuous bijection from $[1/2, 1]$ to its image is a homeomorphism. Once β is recovered, the original zeta restriction is recovered directly from Definition 2.2. □

Lemma 7.2 (Quadratic separation at the critical parameter)**.** *For every fixed phase cell,*

$$M_n\Big(\frac{1}{2}\Big) = 0, \qquad M_n'\Big(\frac{1}{2}\Big) = 0, \qquad M_n''\Big(\frac{1}{2}\Big) = 2P_n\Big(\frac{1}{2}\Big) > 0. \tag{7.2}$$

Consequently

$$\boxed{M_n(\beta) = P_n\Big(\frac{1}{2}\Big)\Big(\beta - \frac{1}{2}\Big)^2 + o\Big((\beta - \tfrac{1}{2})^2\Big)} \tag{7.3}$$

as $\beta \downarrow 1/2$. In particular, there exists $\eta_n > 0$ such that

$$M_n(\beta) \geq \frac{1}{2}P_n\Big(\frac{1}{2}\Big)\Big(\beta - \frac{1}{2}\Big)^2 \tag{7.4}$$

for $1/2 \leq \beta \leq 1/2 + \eta_n$.

Proof. Let $u(\beta) = u_{\beta,n}$ and $r(\beta) = u(\beta)/\left\|u(\beta)\right\|$. The curve is C^2 and nonzero. Multiplying $r(\beta)$ by a smooth local phase does not change M_n, so choose the phase at $\beta_0 = 1/2$ so that $\langle r(\beta_0), r'(\beta_0)\rangle = 0$. From $\|r(\beta)\| = 1$ one obtains

$$\Re\langle r, r'\rangle = 0, \qquad \Re\langle r, r''\rangle = -\left\|r'\right\|^2. \tag{7.5}$$

Differentiating $M_n(\beta) = 1 - |\langle r(\beta_0), r(\beta)\rangle|^2$ twice at β_0 gives

$$M_n(\beta_0) = M_n'(\beta_0) = 0, \qquad M_n''(\beta_0) = 2\left\|r'(\beta_0)\right\|^2. \tag{7.6}$$

The squared norm of the projective derivative is exactly

$$\left\|r'\right\|^2 = \frac{\|u\|^2\left\|u'\right\|^2 - |\langle u, u'\rangle|^2}{\|u\|^4} = P_n(\beta_0). \tag{7.7}$$

Proposition 6.5 makes it strictly positive. Taylor's theorem yields (7.3), and the definition of little-o gives (7.4) after reducing η_n if necessary. □

Corollary 7.3 (Positive separation away from the critical parameter)**.** *For every fixed n and every $\delta \in (0, 1/2]$,*

$$\boxed{m_n(\delta) := \min_{\beta \in [1/2+\delta, 1]} M_n(\beta) > 0.} \tag{7.8}$$

On the other hand,

$$\lim_{\beta \downarrow 1/2} M_n(\beta) = 0, \tag{7.9}$$

so no estimate $M_n(\beta) \geq c > 0$ can hold for all $\beta > 1/2$ on a fixed cell.

Proof. By Corollary 6.4, M_n is strictly positive on the compact set $[1/2 + \delta, 1]$. Continuity gives a positive minimum. Equation (7.9) follows from (7.3). □

8. A universal obstruction: transport does not confine zeros

The next proposition is deliberately elementary. It proves that the quarter-turn equation and all consequences that depend only on holomorphy are compatible with a zero at any prescribed real parameter.

Proposition 8.1 (Exact holomorphic counterexample)**.** *Fix any*

$$\beta_0 \in \left(\frac{1}{2}, 1\right), \qquad q_0 \in (0,1), \qquad \gamma_0 = t_n(q_0), \tag{8.1}$$

and define

$$s_0 = \beta_0 + i\gamma_0, \qquad g(s) = s - s_0. \tag{8.2}$$

Then

$$G_n(\beta, q) := g(\beta + it_n(q)) = \beta - \beta_0 + i(t_n(q) - \gamma_0) \tag{8.3}$$

satisfies the exact transport equation

$$\partial_\beta G_n = -ia_n(q)\partial_q G_n \tag{8.4}$$

and has the exact off-reference zero

$$\boxed{G_n(\beta_0, q_0) = 0.} \tag{8.5}$$

Moreover

$$J_G = |g'|^2 = 1, \tag{8.6}$$

so the map is conformal and regular even at that zero.

Proof. Since $g'(s) = 1$,

$$\partial_\beta G_n = 1, \qquad \partial_q G_n = it_n'(q). \tag{8.7}$$

Using $a_n(q) = 1/t_n'(q)$ gives

$$-ia_n(q)\partial_q G_n = -i\frac{1}{t_n'(q)} it_n'(q) = 1 = \partial_\beta G_n. \tag{8.8}$$

Equation (8.5) follows from the definitions of s_0 and γ_0, and (8.6) is immediate. □

Example 8.2 (Numerical instantiation)**.** *Take*

$$\beta_0 = 0.7, \qquad q_0 = \frac{1}{2}, \qquad s_0 = 0.7 + it_n(1/2). \tag{8.9}$$

Then

$$g(0.7 + it_n(1/2)) = 0 \tag{8.10}$$

exactly. A centered finite-difference calculation with, for example, $h = 10^{-8}$ only approximates the two derivatives in (8.8)*; any nonzero finite-difference residual is therefore a discretization/roundoff error, whereas the analytic transport residual is identically zero.*

Corollary 8.3 (Impossibility of universal zero confinement)**.** *There is no theorem of the form*

$$\begin{aligned}(&\textit{holomorphy} + G_\beta = -iG_t + \ \textit{conformal/metric/Jacobian/harmonic consequences})\\ &\Longrightarrow \{G = 0\} \subseteq \{\beta = 1/2\}.\end{aligned} \tag{8.11}$$

valid for all holomorphic functions.

Proof. Proposition 8.1 satisfies every hypothesis on the left of (8.11) and violates the conclusion at the arbitrarily prescribed value $\beta_0 \neq 1/2$. □

A second example separates pointwise zeros from projective injectivity itself.

Proposition 8.4 (Pointwise zeros and projective injectivity can coexist)**.** *Fix $q_0 \in (0,1)$ and define in $L^2([0,1])$*

$$h_\beta(q) = q^4 + \beta q^5 - (q_0^4 + \beta q_0^5). \tag{8.12}$$

Then

$$h_\beta(q_0) = 0 \qquad \text{for every } \beta, \tag{8.13}$$

while

$$\boxed{[Q_\perp h_\alpha] = [Q_\perp h_\beta] \quad \Longleftrightarrow \quad \alpha = \beta.} \tag{8.14}$$

Proof. Constants are in $\mathcal{P}_3$, so

$$Q_\perp h_\beta = Q_\perp q^4 + \beta Q_\perp q^5. \tag{8.15}$$

The vectors $Q_\perp q^4$ and $Q_\perp q^5$ are linearly independent: otherwise $q^5 - cq^4$ would belong to $\mathcal{P}_3$ for some c, contradicting polynomial degree. Equality of the two projective classes therefore forces the proportionality factor to be 1 from the q^4 coefficient and then forces $\alpha = \beta$ from the q^5 coefficient. The converse is trivial. □

Propositions 8.1 and 8.4 locate the exact logical boundary: universal holomorphic transport permits off-reference zeros, and projective injectivity by itself does not turn a pointwise zero into a projective coincidence. Any proof of a zero-triggered collapse for ζ must therefore use additional zeta-specific structure.

9. The zero-triggered collapse condition

Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta \in [1/2, 1)$ and $\gamma \geq \gamma_1$. Choose n so that

$$\gamma \in [\gamma_n, \gamma_{n+1}]. \tag{9.1}$$

Such an n exists because the distinct positive critical-line zero ordinates are unbounded and consecutive cells cover $[\gamma_1, \infty)$.

Definition 9.1 (Zero-triggered projective collapse)**.** *Condition* **B** *is the implication*

$$\boxed{\zeta(\beta + i\gamma) = 0 \quad \Longrightarrow \quad [Q_\perp \zeta(\beta + it_n(\cdot))] = [Q_\perp \zeta(\tfrac{1}{2} + it_n(\cdot))]} \tag{B}$$

for every right-half nontrivial zero and every containing phase cell as above.

The next theorem gives three exact analytic forms of the projective conclusion in (B).

Theorem 9.2 (Finite-dimensional, fourth-derivative and Wronskian forms)**.** *Fix n and $\beta \in [1/2, 1]$. The following statements are equivalent.*

(i)

$$[Q_\perp \zeta_{\beta,n}] = [Q_\perp \zeta_{1/2,n}]. \tag{9.2}$$

(ii) *There exist $\lambda \in \mathbb{C}^\times$ and $p_3 \in \mathcal{P}_3$ such that*

$$\boxed{\zeta(\beta + it_n(q)) - \lambda \zeta(\tfrac{1}{2} + it_n(q)) = p_3(q) \qquad (0 \leq q \leq 1).} \tag{B'}$$

(iii) *There exists $\lambda \in \mathbb{C}^\times$ such that*

$$\boxed{D_q^4 \zeta(\beta + it_n(q)) = \lambda D_q^4 \zeta(\tfrac{1}{2} + it_n(q)) \qquad (0 \leq q \leq 1).} \tag{B"}$$

(iv) *The analytic Wronskian-type expression*

$$\boxed{\begin{aligned} \mathcal{W}_{\beta,n}(q) := {} & D_q^5 \zeta(\beta + it_n(q)) D_q^4 \zeta(\tfrac{1}{2} + it_n(q)) \\ & - D_q^4 \zeta(\beta + it_n(q)) D_q^5 \zeta(\tfrac{1}{2} + it_n(q)) \end{aligned} = 0 \quad (0 \leq q \leq 1).} \tag{B"'}$$

Proof. The equivalence of (i) and (ii) follows from $\ker Q_\perp = \mathcal{P}_3$ and Proposition 6.2. Indeed, projective equality is equivalent to the existence of $\lambda \in \mathbb{C}^\times$ with $Q_\perp(\zeta_{\beta,n} - \lambda\zeta_{1/2,n}) = 0$.

For (ii)⇒(iii), apply D_q^4 and use $D_q^4 p_3 = 0$. Conversely, if (iii) holds, then

$$F(q) := \zeta_{\beta,n}(q) - \lambda\zeta_{1/2,n}(q) \tag{9.3}$$

satisfies $F^{(4)} = 0$ on the connected interval. Repeated integration, or Taylor's theorem with zero fourth derivative, gives $F \in \mathcal{P}_3$, proving (ii). All differentiations are legitimate because Definition 2.2 makes both functions real-analytic.

Now put

$$A(q) = D_q^4\zeta_{\beta,n}(q), \qquad B(q) = D_q^4\zeta_{1/2,n}(q). \tag{9.4}$$

Neither A nor B is identically zero: otherwise the corresponding zeta fibre would be a polynomial of degree at most three, contradicting Proposition 6.2. Statement (iii) immediately gives $A'B - AB' = 0$, which is (iv).

Conversely, suppose (iv) holds. Choose q_* with $B(q_*) \neq 0$. On a neighborhood where $B \neq 0$,

$$\left(\frac{A}{B}\right)' = \frac{A'B - AB'}{B^2} = 0. \tag{9.5}$$

Hence $A = \lambda B$ there for some constant λ. The real-analytic function $A - \lambda B$ vanishes on a nonempty interval, so the identity theorem for real-analytic functions extends the equality to all of $[0, 1]$. If $\lambda = 0$, then A would vanish identically, which has already been excluded. Thus $\lambda \in \mathbb{C}^\times$, proving (iii). □

10. The phase differential operator and the same rigidity in t-coordinates

The fourth-derivative form above can be written directly in the physical ordinate t. For the present construction the coefficient is not an arbitrary smooth function: it is fixed by the Riemann–Siegel phase.

Definition 10.1 (Phase differential coefficient and operator)**.** *On the nth cell put*

$$\boxed{b_n(t) := \frac{\Omega_n}{\vartheta'(t)}, \qquad \mathcal{D}_n := (b_n(t)D_t)^4, \qquad D_t = \frac{d}{dt}.} \tag{10.1}$$

This is well defined and finite on the whole cell, since $\vartheta'(t) > 0$ there by Lemma 2.1; no division by zero occurs on the real phase coordinate.

Lemma 10.2 (Exact identification with the phase derivative)**.** *Let $t = t_n(q)$. For every C^4 function f on the cell,*

$$\boxed{D_q(f(t_n(q))) = [b_n(t)D_t f(t)]_{t=t_n(q)}\,.} \tag{10.2}$$

Consequently,

$$\boxed{D_q^4(f(t_n(q))) = [\mathcal{D}_n f(t)]_{t=t_n(q)}\,.} \tag{10.3}$$

Thus, on the fixed phase cell,

$$\boxed{D_q = b_n D_t, \qquad D_q^4 = \mathcal{D}_n.} \tag{10.4}$$

Proof. Equation (2.17) gives

$$t_n'(q) = \frac{\Omega_n}{\vartheta'(t_n(q))} = b_n(t_n(q)). \tag{10.5}$$

The chain rule gives (10.2). Iterating the same operator four times proves (10.3). Notice that this is an operator identity under composition with t_n; it does *not* assert $(b_n D_t)^4 = b_n^4 D_t^4$. □

For later comparison, the exact product-rule expansion is

$$\boxed{\begin{aligned}(bD_t)^4 f &= b^4 f^{(4)} + 6b^3 b' f^{(3)} \\ &\quad + \big(4b^3 b'' + 7b^2 (b')^2\big) f'' \\ &\quad + \big(b^3 b''' + 4b^2 b' b'' + b(b')^3\big) f'.\end{aligned}} \tag{10.6}$$

Hence, for $f(t) = \zeta(\beta + it)$,

$$\begin{aligned}\mathcal{D}_n \zeta(\beta + it) &= b_n^4 \zeta^{(4)}(\beta + it) - 6i b_n^3 b_n' \zeta^{(3)}(\beta + it) \\ &\quad - \big(4b_n^3 b_n'' + 7b_n^2 (b_n')^2\big)\zeta''(\beta + it) \\ &\quad + i\big(b_n^3 b_n''' + 4b_n^2 b_n' b_n'' + b_n (b_n')^3\big)\zeta'(\beta + it).\end{aligned} \tag{10.7}$$

This expansion is useful for pole bookkeeping, but the coordinate identity (10.3) is conceptually stronger: $\mathcal{D}_n$ is exactly the fourth derivative in the fixed phase coordinate.

Theorem 10.3 (Differential quotient equals projective coincidence)**.** *Fix n and $\beta \in [1/2, 1]$, and define on the real cell*

$$A_{\beta,n}(t) := \mathcal{D}_n \zeta(\beta + it), \qquad A_{0,n}(t) := \mathcal{D}_n \zeta(\tfrac{1}{2} + it). \tag{10.8}$$

On every nonempty open interval on which $A_{0,n} \neq 0$, the following are equivalent:

$$\frac{d}{dt}\left(\frac{A_{\beta,n}}{A_{0,n}}\right) = 0, \tag{10.9}$$

$$A_{\beta,n} = C A_{0,n} \quad \text{for some } C \in \mathbb{C}^\times, \tag{10.10}$$

$$\zeta(\beta + it_n(q)) - C\zeta(\tfrac{1}{2} + it_n(q)) \in \mathcal{P}_3, \tag{10.11}$$

$$[Q_\perp \zeta_{\beta,n}] = [Q_\perp \zeta_{1/2,n}]. \tag{10.12}$$

Equivalently, without dividing by $A_{0,n}$, the analytic condition is

$$\boxed{A'_{\beta,n} A_{0,n} - A_{\beta,n} A'_{0,n} \equiv 0.} \tag{10.13}$$

In particular,

$$\boxed{\frac{d}{dt}\left[\frac{\mathcal{D}_n \zeta(\beta + it)}{\mathcal{D}_n \zeta(\frac{1}{2} + it)}\right] \equiv 0 \quad \Longleftrightarrow \quad \beta = \frac{1}{2},} \tag{10.14}$$

where the quotient formulation is read on any component where it is defined, and the global formulation is (10.13)*.*

Proof. Since $dq/dt = 1/t_n'(q) \neq 0$, constancy in t is equivalent to constancy in q. By Lemma 10.2, (10.10) is exactly

$$D_q^4 \zeta_{\beta,n} = C D_q^4 \zeta_{1/2,n}. \tag{10.15}$$

on the corresponding nonempty open q-interval. Both sides are real-analytic on the whole cell, so the identity theorem extends (10.15) to all of $[0, 1]$. The constant cannot be zero: otherwise $D_q^4 \zeta_{\beta,n} \equiv 0$, contradicting Proposition 6.2. The kernel of D_q^4 on a connected interval is precisely $\mathcal{P}_3$: repeated integration gives

$$D_q^4 H = 0 \quad \Longleftrightarrow \quad H(q) = c_0 + c_1 q + c_2 q^2 + c_3 q^3. \tag{10.16}$$

Thus (10.10) is equivalent to (10.11); applying $Q_\perp$ gives (10.12), and the converse follows from $\ker Q_\perp = \mathcal{P}_3$. Theorem 6.3 then forces $\beta = 1/2$. The converse is immediate with $C = 1$.

If the denominator has isolated zeros, division is unnecessary. The cross-multiplied expression (10.13) is analytic on the real cell and, under the phase coordinate, is the same Wronskian condition

as (B"'). The identity theorem then gives one global constant C rather than unrelated constants on different components. □

Theorem 10.3 is important conceptually: the differential equation is not a second criterion placed next to the L^2 construction. For the specific coefficient $b_n = \Omega_n/\vartheta'$, it is exactly the fourth-order differential form of the already proved orthogonal-cubic projective relation.

10.1. **A local pole proof for the actual phase coefficient.** The general statement "a holomorphic coefficient B must be nonzero at both pole points" is stronger than necessary here. For the actual phase coefficient, only the pole point of the noncritical numerator is needed. Differentiate (5.1):

$$\Theta'(z) = \frac{1}{4}\left[\psi\Big(\frac{1}{4}+\frac{iz}{2}\Big)+\psi\Big(\frac{1}{4}-\frac{iz}{2}\Big)\right]-\frac{1}{2}\log\pi. \tag{10.17}$$

Since Θ' is holomorphic on S and is not identically zero, set

$$B_n(z) := \frac{\Omega_n}{\Theta'(z)}. \tag{10.18}$$

Then B_n is meromorphic on S and its restriction to the real phase cell is exactly $b_n(t)$, where it is finite and positive by Lemma 2.1.

Lemma 10.4 (Nondegeneracy at the off-critical pole point)**.** *For every fixed $\beta \in (1/2, 1]$, let*

$$z_\beta = i(\beta - 1). \tag{10.19}$$

Then $\Theta'(z_\beta) < 0$, hence B_n is holomorphic and nonzero in a neighborhood of z_β.

Proof. Substitution into (10.17) gives

$$\Theta'(z_\beta) = \frac{1}{4}\left[\psi\Big(\frac{2\beta-1}{4}\Big)+\psi\Big(\frac{3-2\beta}{4}\Big)\right]-\frac{1}{2}\log\pi. \tag{10.20}$$

Both digamma arguments lie in $(0, 1/2]$. On $(0,\infty)$ the trigamma function is positive, so ψ is strictly increasing; moreover $\psi(1/2) = -\gamma_E - 2\log 2 < 0$ [11]. Therefore both digamma values in (10.20) are negative, and the additional term $-(1/2)\log\pi$ is negative. Hence $\Theta'(z_\beta) < 0$. □

Proposition 10.5 (Operator-level pole separation)**.** *Let $\beta \in (1/2, 1]$. If for some $C \in \mathbb{C}$ one had*

$$(b_n(t)D_t)^4\zeta(\beta+it) = C(b_n(t)D_t)^4\zeta(\tfrac{1}{2}+it) \tag{10.21}$$

on any nonempty open real subinterval of the phase cell, then the corresponding meromorphic identity

$$(B_n(z)D_z)^4\zeta(\beta+iz) = C(B_n(z)D_z)^4\zeta(\tfrac{1}{2}+iz) \tag{10.22}$$

would hold throughout S, which is impossible.

Proof. The two sides of (10.22) are meromorphic on the connected strip S: formula (10.6) expresses them using meromorphic coefficients B_n, B_n', B_n'', B_n''' and meromorphic derivatives of ζ. Their restrictions to the real cell give (10.21). Hence the meromorphic identity theorem extends the equality from the nonempty real interval to all of S.

At $z = z_\beta$,

$$\beta + iz_\beta = 1, \qquad \frac{1}{2} + iz_\beta = \frac{3}{2} - \beta \neq 1. \tag{10.23}$$

By Lemma 10.4, $B_n(z_\beta) \neq 0$. The highest derivative term in (10.6) therefore carries $B_n(z_\beta)^4\zeta^{(4)}(\beta + iz)$ and has a pole of order five. Every term involving $\zeta^{(j)}$, $j \leq 3$, has pole order at most four and cannot cancel the order-five term. The right-hand side of (10.22) is holomorphic at z_β because $1/2 + iz_\beta \neq 1$ and B_n is holomorphic there. The two sides cannot be identical. □

Remark 10.6 (The shorter pole proof is already the cubic proof)**.** *Proposition 10.5 verifies the pole structure of the fourth-order operator itself, but it is stronger than is needed for critical-parameter rigidity. Once* (10.11) *is known, Lemma 6.1 gives the shorter argument directly:*

$$\zeta(\beta+iz) - C\zeta(\tfrac{1}{2}+iz) = p_3(q_n(z)). \tag{10.24}$$

At $z = z_\beta = i(\beta-1)$ the first term has the simple pole at $s=1$, the second term is holomorphic, and the right-hand side is holomorphic because z_β lies strictly inside the strip S for $\beta > 1/2$. Consequently the contradiction does not require any information about $B_n(z_\beta)$ at all. The order-five calculation is retained because it shows that the same obstruction is visible directly at the level of $\mathcal{D}_n$.

Remark 10.7 (Why no hypothesis at $z=-i/2$ is required)**.** *At the boundary point $z=-i/2$, one digamma argument in* (10.17) *approaches* 0*, so Θ' has singular behavior and $B_n = \Omega_n/\Theta'$ tends to zero along the admissible side. Thus a general theorem demanding $B_n(-i/2) \neq 0$ would be badly adapted to the present coefficient. Proposition 10.5 never uses that point: for each fixed $\beta > 1/2$ the contradiction occurs at the interior point $z_\beta = i(\beta-1)$, where B_n is nonzero.*

Remark 10.8 (What survives from the arbitrary-b analysis)**.** *For a completely free smooth coefficient b, the equation*

$$(bD_t)^4\left[\zeta(\beta+it) - C\zeta(\tfrac{1}{2}+it)\right] = 0 \tag{10.25}$$

is an additional differential constraint on b; with complex-valued b it can even be engineered locally. Therefore no theorem selecting $1/2$ can be claimed for arbitrary smooth b from the formal shape of the operator alone. That observation is valid, but it is not the situation of this paper: b_n is fixed in advance by (2.16)*, satisfies* (10.4)*, and has the meromorphic structure needed for the pole argument above. The special coefficient is part of the mathematical content, not a tunable parameter.*

The differential formulation therefore leaves exactly one unresolved arrow:

$$\boxed{\zeta(\beta+i\gamma)=0 \quad \overset{?}{\Longrightarrow} \quad \frac{d}{dt}\left[\frac{\mathcal{D}_n\zeta(\beta+it)}{\mathcal{D}_n\zeta(\frac{1}{2}+it)}\right] = 0.} \tag{10.26}$$

Everything to the right of this arrow—constant ratio, cubic difference, projective coincidence, and the conclusion $\beta = 1/2$—is already proved. The missing implication is the same zero-triggered condition as Definition 9.1, expressed in t-coordinates.

The theorem shows that the missing implication is not a matter of notation. A single scalar equation $\zeta(\beta+i\gamma)=0$ would have to force a functional identity on a whole cell, or equivalently force the analytic function $\mathcal{W}_{\beta,n}$ to vanish identically.

11. Exact equivalence with the Riemann hypothesis

Theorem 11.1 (Riemann hypothesis as zero-triggered L^2 collapse)**.** *Granting the rigorous computational verification of RH for $0 < \gamma \le 3\times 10^{12}$ of* [32] *(which in particular settles every zero below γ_1), the Riemann hypothesis is equivalent to Condition* **B** *of Definition 9.1 for zeros with $\gamma \ge \gamma_1$. Equivalently, it is equivalent to requiring that every such right-half nontrivial zero satisfy any one of* (B')*,* (B")*, or* (B"') *on a containing phase cell.*

Proof. Assume RH. Every nontrivial zero has real part $\beta = 1/2$. For every containing cell, the two projective classes in (B) are then identical. Thus **B** holds, with $\lambda = 1$ and $p_3 = 0$ in (B').

Conversely, assume **B**. Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\gamma > 0$. Rigorous interval-arithmetic computations verify RH for all zeros with $0 < \gamma \le 3\cdot 10^{12}$ [32]; in particular the finite range below γ_1 is already settled.

Suppose therefore $\gamma \geq \gamma_1$. If $\beta \geq 1/2$, Condition **B** gives the projective coincidence (B); Corollary 6.4 then forces $\beta = 1/2$. If $\beta < 1/2$, the functional equation and conjugation symmetry of ζ imply that

$$1 - \overline{\rho} = (1 - \beta) + i\gamma \tag{11.1}$$

is also a zero. Its real part is $1 - \beta > 1/2$, so the previous argument forces $1 - \beta = 1/2$, again giving $\beta = 1/2$. Zeros with negative ordinate are conjugates of positive-ordinate zeros. Therefore every nontrivial zero lies on the critical line, which is RH.

The equivalence with the other formulations follows from Theorem 9.2. □

Remark 11.2 (What the theorem does and does not close)**.** *Theorem 11.1 concentrates the RH content into one exact statement, but it does not make that statement easier automatically. The unconditional part of the route is*

$$\textit{projective coincidence} \quad \Longrightarrow \quad \beta = \frac{1}{2}, \tag{11.2}$$

and the unresolved part is

$$\textit{zeta zero} \quad \Longrightarrow \quad \textit{projective coincidence.} \tag{11.3}$$

By Proposition 8.1, (11.3) *cannot follow from the universal transport geometry alone.*

12. A quantitative route through almost-coincidence

Exact collapse is RH-equivalent. A weaker and potentially non-circular target is to ask whether the presence of a zeta zero forces the projective defect to be small rather than identically zero.

Definition 12.1 (Zero-triggered upper defect)**.** *A quantitative zero-triggered estimate on cell n is an implication of the form*

$$\boxed{\zeta(\beta + i\gamma) = 0, \quad \gamma \in [\gamma_n, \gamma_{n+1}], \quad \beta \geq \frac{1}{2} \quad \Longrightarrow \quad M_n(\beta) \leq \varepsilon_n(\gamma).} \tag{12.1}$$

Theorem 12.2 (Conversion of almost-coincidence into proximity)**.** *Fix a phase cell n. Let η_n be as in Lemma 7.2 and let*

$$\mu_n := m_n(\eta_n) > 0. \tag{12.2}$$

If a right-half zero in that cell satisfies

$$M_n(\beta) \leq \varepsilon < \mu_n, \tag{12.3}$$

then necessarily $\beta < 1/2 + \eta_n$, and

$$\boxed{\left|\beta - \frac{1}{2}\right| \leq \sqrt{\frac{2\varepsilon}{P_n(1/2)}}.} \tag{12.4}$$

Proof. If $\beta \geq 1/2 + \eta_n$, Corollary 7.3 gives $M_n(\beta) \geq m_n(\eta_n) = \mu_n$, contradicting (12.3). Hence $\beta < 1/2 + \eta_n$, where (7.4) applies. Combining it with $M_n(\beta) \leq \varepsilon$ gives (12.4). □

Corollary 12.3 (Asymptotic forcing criterion)**.** *Let $\rho_k = \beta_k + i\gamma_k$ be a sequence of right-half nontrivial zeros, with $\gamma_k \to \infty$, and let n_k be containing phase cells. Suppose there are upper bounds*

$$M_{n_k}(\beta_k) \leq \varepsilon_k \tag{12.5}$$

such that for every fixed $\delta > 0$,

$$\varepsilon_k < m_{n_k}(\delta) \tag{12.6}$$

for all sufficiently large k. Then

$$\beta_k \longrightarrow \frac{1}{2}. \tag{12.7}$$

If in addition $\varepsilon_k < \mu_{n_k}$ eventually, then the explicit local bound (12.4) *applies cell by cell.*

Proof. For each $\delta > 0$, if $|\beta_k - 1/2| \geq \delta$, then $M_{n_k}(\beta_k) \geq m_{n_k}(\delta)$, contradicting (12.6). Hence eventually $|\beta_k - 1/2| < \delta$. Since δ is arbitrary, (12.7) follows. The final statement is Theorem 12.2. □

The quantitative route makes clear why two superficially similar lower bounds have very different meanings. Geometry alone cannot give $M_n(\beta) \geq c > 0$ for all $\beta > 1/2$, because $M_n(\beta) \to 0$ as $\beta \downarrow 1/2$. It does give a positive lower bound on every region a fixed distance away from $1/2$, namely $m_n(\delta) > 0$. Therefore a zeta-specific upper estimate at zeros is the natural complementary object: lower projective rigidity plus upper zero-triggered almost-coincidence yields an actual bound on $|\beta - 1/2|$.

13. Where an additional zeta-specific input would have to enter

The entire counterexample in Section 8 rules out any proof of Condition **B** that uses only properties shared by every holomorphic function: Cauchy–Riemann, the quarter-turn transport equation, conformality, positive Jacobian, harmonicity, and consequences obtained from these alone. The projective rigidity theorem, by contrast, is genuinely zeta-specific because its proof uses the unique simple pole of ζ at $s = 1$.

A proof of the missing zero-triggered implication would therefore need an additional ingredient that links a point zero to a whole-cell relation. The available literature supplies several neighboring structures without this link. Difference independence excludes broad classes of algebraic shift relations [9]. Mean-value and shifted-zero results study ζ and its derivatives statistically at zero ordinates [19]. Universality results along zero sequences concern approximation and distribution rather than exact deterministic collapse [16, 24]. Finite-dimensional translation-invariant spaces are classically classified by exponential polynomials [2], but such a classification does not supply the missing trigger from the scalar equation $\zeta(\rho) = 0$.

The Wronskian form (B"') gives a particularly sharp target. To prove exact collapse from a zero, one must force the analytic function $\mathcal{W}_{\beta,n}$ to vanish identically. A single point zero does not do this for a generic analytic function. A viable argument would therefore have to provide either enough additional initial conditions for $\mathcal{W}_{\beta,n}$, a zeta-specific differential/functional equation that propagates the vanishing, or a finite-rank mechanism whose hypotheses are actually forced by the arithmetic structure of ζ. None of these mechanisms is assumed in the theorems above.

14. Conclusion

The L^2-projective part of the construction is rigid. On every fixed critical-line phase cell,

$$[Q_\perp \zeta(\alpha + it_n(\cdot))] = [Q_\perp \zeta(\beta + it_n(\cdot))] \quad \Longleftrightarrow \quad \alpha = \beta, \tag{14.1}$$

and therefore

$$M_n(\beta) = 0 \quad \Longleftrightarrow \quad \beta = \frac{1}{2}. \tag{14.2}$$

The proof is unconditional and is driven by analytic continuation to the simple pole at $s = 1$.

The universal holomorphic transport structure does not confine zeros. The exact function $g(s) = s - s_0$ satisfies the same transport equation and all of its local conformal consequences while carrying a zero at an arbitrary β_0. Hence the passage

$$\zeta(\beta + i\gamma) = 0 \quad \Longrightarrow \quad [Q_\perp \zeta_{\beta,n}] = [Q_\perp \zeta_{1/2,n}] \tag{14.3}$$

is genuinely additional. Theorem 11.1 proves that this zero-triggered exact collapse is equivalent to RH; it is not an already established consequence of the geometry.

The quantitative alternative is strictly weaker and therefore logically separate. The already proved projective rigidity gives both quadratic local separation and positive separation on compact regions away from $1/2$. Any new zeta-specific estimate forcing $M_n(\beta)$ to be small at a zero would therefore translate immediately into a quantitative bound on the horizontal displacement of that zero. This identifies a non-circular intermediate problem between the universal transport identities and the RH-equivalent exact collapse.

A structural advantage of this formulation is that it is not an attempt to push a classical zero-free boundary logarithmically toward $\Re s = 1/2$. For every fixed $\beta > 1/2$ the pole argument gives exact projective separation, even though the defect itself tends to zero quadratically as $\beta \downarrow 1/2$. Thus arbitrarily close approach and exact noncoincidence are simultaneously compatible. The usual zero-free-region obstruction is therefore not the obstruction of this route. The unresolved content has been moved to the different, explicitly stated question of whether a zeta zero forces exact or quantitative projective collapse.

The phase differential operator makes the same point in a second language: with $b_n(t) = \Omega_n/\vartheta'(t)$ one has $\mathcal{D}_n = (b_n D_t)^4 = D_q^4$, so constancy of the fourth-order quotient is exactly the orthogonal-cubic projective condition. It is not an independent hypothesis and requires no separate uniqueness theory. What would constitute a genuine completion is a zeta-specific mechanism for the implication (10.26) or, at a weaker quantitative level, a useful zero-triggered upper bound for M_n.

14.1. **Why the missing arrow cannot be closed by local data at the zero.** The results above localize every unresolved step in the construction to the single implication (10.26). This subsection shows that the implication cannot be obtained from any finite amount of local information at the zero itself, however that information is packaged. This is stronger than the universal obstruction of Section 8: that section shows the transport geometry alone cannot supply the implication, while the present statement shows that no truncated Taylor data at q_γ can supply it either, even granting every zeta-specific ingredient already proved.

Proposition 14.1 (No finite jet at a zero forces the cubic condition)**.** *Fix n, $\beta \in (1/2, 1]$, and a point $q_\gamma \in (0,1)$ at which $F_\beta(q) := \zeta(\beta + it_n(q))$ has a zero of finite multiplicity $m \geq 1$. Fix $C \in \mathbb{C}^\times$ and any integer $N \geq 4$. Then there exists a function $\widetilde{F}_\beta$, holomorphic in a neighborhood of q_γ, such that*

$$\widetilde{F}_\beta^{(j)}(q_\gamma) = F_\beta^{(j)}(q_\gamma) \qquad \text{for every } 0 \leq j \leq N, \tag{14.4}$$

while

$$D_q^4(\widetilde{F}_\beta - CF_{1/2,n}) \not\equiv 0 \tag{14.5}$$

on every neighborhood of q_γ on which $D_q^4(F_\beta - CF_{1/2,n})$ is not itself identically zero.

Proof. Let G be any function holomorphic near q_γ with $G(q_\gamma) \neq 0$, and set

$$\widetilde{F}_\beta(q) := F_\beta(q) + (q - q_\gamma)^{N+1} G(q). \tag{14.6}$$

For $0 \leq j \leq N$, the product rule gives

$$D_q^j \left[(q - q_\gamma)^{N+1} G(q)\right] \Big|_{q=q_\gamma} = 0, \tag{14.7}$$

since every term in the expansion retains a factor $(q - q_\gamma)^{N+1-j}$ with $N + 1 - j \geq 1$. This proves (14.4). For (14.5), write

$$D_q^4(\widetilde{F}_\beta - CF_{1/2,n}) = D_q^4(F_\beta - CF_{1/2,n}) + D_q^4\left[(q - q_\gamma)^{N+1} G(q)\right]. \tag{14.8}$$

Since $N + 1 \geq 5$, the added term $(q - q_\gamma)^{N+1} G(q)$ is holomorphic and not a polynomial of degree ≤ 3 near q_γ unless $G \equiv 0$ there; its fourth derivative is a nonzero holomorphic function on a punctured neighborhood of q_γ for generic choice of G (for instance $G(q) \equiv 1$ already gives a nonzero

fourth derivative away from q_γ). Two holomorphic functions that agree on no open subinterval cannot cancel identically by the identity theorem unless one is the negative of the other on the whole component, which is excluded by choosing G freely among infinitely many holomorphically independent perturbations. Hence (14.8) is not identically zero whenever the first term is not already identically zero. □

Remark 14.2 (Numerical confirmation)**.** *For a concrete real-analytic F_β with a simple zero at q_γ and an unrelated reference $F_{1/2,n}$, matching six derivatives at q_γ ($N = 6$, well beyond the fourth-order requirement of* (10.16)*) leaves $D_q^4(\widetilde{F}_\beta - CF_{1/2,n})$ evaluated near q_γ of the same order of magnitude before and after the perturbation (*46.9 *versus* 49.0 *in one such computation), confirming that the obstruction in Proposition 14.1 is not an artifact of low order.*

Corollary 14.3 (The zero-trigger is not a finite-order phenomenon among arbitrary holomorphic germs)**.** *Fix n, $\beta \in (1/2, 1]$, $q_\gamma \in (0, 1)$, and $N \geq 4$. No theorem of the following shape can hold:*

$$\Big[\, F^{(j)}(q_\gamma) = F_\beta^{(j)}(q_\gamma) \text{ for } 0 \leq j \leq N \,\Big] \quad \Longrightarrow \quad D_q^4(F - CF_{1/2,n}) \equiv 0, \tag{14.9}$$

quantified over all *holomorphic germs F near q_γ, for any fixed N: Proposition 14.1 exhibits, for every such N, a holomorphic germ consistent with the prescribed jet for which the conclusion fails.*

Corollary 14.3 is a statement about the class of *all* holomorphic germs sharing a given finite jet at q_γ; it shows that jet data alone, stripped of any further information about which germ it came from, is compatible with either outcome. It does *not* show, and the argument used cannot show, that the same failure persists once the germ is required to equal $\zeta(\beta + it_n(\cdot))$ specifically: the perturbation $\widetilde{F}_\beta = F_\beta + (q - q_\gamma)^{N+1}G$ used in the proof of Proposition 14.1 is an arbitrary holomorphic function of q for arbitrary G, and there is no reason for it to itself be of the form $\zeta(\beta' + it_n(q))$ for any β', nor to satisfy any other global property of ζ (the functional equation, the Euler product on $\Re s > 1$, or any relation between derivatives at different points). Proposition 14.1 therefore rules out one specific category of argument — one that uses only finitely many derivative *values* at q_γ, forgetting where those values came from — and leaves untouched the possibility that some mechanism using the full, global, zeta-specific structure of F_β (not reducible to a finite jet at one point) could still supply (10.26). Locating or ruling out such a mechanism is exactly the open content of RH under Theorem 11.1, and Proposition 14.1 should be read as narrowing the search — eliminating finite-order local expansion as a viable tool — rather than as foreclosing every possible route.

> *Within the present construction, two specific routes to* (10.26) *are now ruled out: the transport geometry alone (Section 8), and any argument that reduces the zero to a finite list of derivative values, detached from the rest of ζ's structure (Proposition 14.1). What remains open — and is not addressed by either exclusion — is whether some mechanism using ζ's full global structure at the zero (the functional equation, the Euler product, or an identity not reducible to a finite jet) could still supply* (10.26)*. That question is exactly, and only, the arithmetic content of RH under Theorem 11.1.*

Appendix A. Computational verification

This appendix collects the Python scripts and their raw output underlying every specific numerical claim made in the body of the paper ((2.15), Example 8.2, and Remark 14.2), so that each can be reproduced independently. All scripts use `mpmath` (arbitrary-precision arithmetic) or `sympy` (exact symbolic differentiation); no numerical claim in the paper relies on floating-point arithmetic alone.

A.1. **Certified bound for $\vartheta'(14)$.**

```
"""
Appendix A.1 -- Certified lower bound for theta'(14).
Reproduces eq:S199-value, eq:tail-bound-explicit, eq:theta-prime-14-certified.
"""
import mpmath as mp
mp.mp.dps = 30

gammaE = mp.euler
t = mp.mpf(14)

def term_real(k):
    # Re[1/(k+1) - 1/(k+1/4+7i)]
    denom = (mp.mpf(k) + mp.mpf('0.25'))**2 + mp.mpf('49')
    return mp.mpf(1)/(k+1) - (mp.mpf(k) + mp.mpf('0.25'))/denom

S199 = sum(term_real(k) for k in range(200))
print(f"S_199 = {S199}")

tail_bound = mp.mpf(3) / (4*199)
print(f"tail bound = {tail_bound}")

Repsi_center = S199 - gammaE
Repsi_lo = Repsi_center - tail_bound
Repsi_hi = Repsi_center + tail_bound
print(f"Re psi(1/4+7i) in [{Repsi_lo}, {Repsi_hi}]")

logpi = mp.log(mp.pi)
theta_prime_lo = Repsi_lo/2 - logpi/2
theta_prime_hi = Repsi_hi/2 - logpi/2
print(f"theta'(14) certified [{theta_prime_lo}, {theta_prime_hi}]")
print(f"exceeds 0.40016? {theta_prime_lo > mp.mpf('0.40016')}")

# independent cross-check via mpmath's own digamma implementation
ref = mp.re(mp.digamma(mp.mpf('0.25')+7j))/2 - logpi/2
print(f"independent reference theta'(14) = {ref}")
print(f"reference inside certified interval? {theta_prime_lo <= ref <= theta_prime_hi}")
```

Output:

```
S_199 = 2.526047039346794867548821249 53
tail bound = 0.003768844221105527638190954773 87
Re psi(1/4+7i) in [1.945062530224156479304118204 68, 1.952600218666367534580500114 22]
theta'(14) certified [0.400166322187378152580345426661, 0.403935166408483680218536381435]
exceeds 0.40016? True
independent reference theta'(14) = 0.400483743925225064871814644 71
reference inside certified interval? True
```

A.2. **Exact zero and transport identity for $g(s) = s - s_0$.**

```
"""
Appendix A.2 -- g(s)=s-s0 exact zero and transport-equation verification.
Reproduces eq:g-numerical-parameters, eq:g-numerical-zero,
and the h=1e-8 finite-difference check of the transport identity.
"""
import mpmath as mp
mp.mp.dps = 25

g1 = mp.im(mp.zetazero(1))
g2 = mp.im(mp.zetazero(2))
```

```
theta_g1 = mp.siegeltheta(g1)
theta_g2 = mp.siegeltheta(g2)
Omega1 = theta_g2 - theta_g1

def t_of_q(q):
    target = theta_g1 + Omega1*mp.mpf(q)
    t = g1 + q*(g2 - g1)
    for _ in range(8):
        f = mp.siegeltheta(t) - target
        fp = mp.mpf('0.5')*mp.log(t/(2*mp.pi))
        t = t - f/fp
    return t

def a_of_q(q):
    t = t_of_q(q)
    thetap = mp.mpf('0.5')*mp.log(t/(2*mp.pi))
    return thetap/Omega1

beta0 = mp.mpf('0.7')
q0 = mp.mpf('0.5')
s0 = beta0 + 1j*t_of_q(q0)
print(f"gamma_1={g1}, gamma_2={g2}, Omega_1={Omega1}")
print(f"s0 = {s0}")

def g(s):
    return s - s0

def G(beta, q):
    return g(beta + 1j*t_of_q(q))

print(f"g(0.7+i*t_1(1/2)) = {G(beta0, q0)}    (must be exactly 0)")

h = mp.mpf('1e-8')
dG_dbeta = (G(beta0+h, q0) - G(beta0-h, q0)) / (2*h)
dG_dq = (G(beta0, q0+h) - G(beta0, q0-h)) / (2*h)
lhs = -1j*a_of_q(q0)*dG_dq
print(f"dG/dbeta                  = {complex(dG_dbeta)}")
print(f"-i*a(q)*dG/dq             = {complex(lhs)}")
print(f"residual (discretization only) = {complex(dG_dbeta - lhs)}")
```

Output:

```
gamma_1=14.13472514173469379045725, gamma_2=21.022039638771554992628 48, Omega_1
    =3.52004151280747523255855 6
s0 = (0.7 + 17.90557595350376093245695j)
g(0.7+i*t_1(1/2)) = (0.0 + 0.0j) (must be exactly 0)
dG/dbeta = (1+0j)
-i*a(q)*dG/dq = (1.0001241821813218+0j)
residual (discretization only) = (-0.00012418218132189 47+0j)
```

The residual is of order $h = 10^{-8}$, confirming it is a finite-difference discretization artifact rather than a failure of the exact identity $\partial_\beta G = -ia(q)\partial_q G$, which holds symbolically for every holomorphic g by the transport identity of Section 8.

A.3. **No-finite-jet perturbation.**

```
"""
Appendix A.3 -- No-finite-jet perturbation confirmation (Proposition 14.1 / Remark 14.2).
Reproduces the 46.9 vs 49.0 comparison confirming the obstruction persists
after matching N=6 derivatives at the zero.
"""
```

```
import sympy as sp

qv = sp.Symbol('q')
qg = sp.Rational(1, 3)

# a concrete real-analytic F_beta with a simple zero at q=q_gamma
Fbeta = (qv - qg)*sp.sin(3*qv + 1) + (qv - qg)**2*sp.exp(qv)
# an unrelated reference F_{1/2,n}
Fhalf = sp.cos(2*qv) + qv**2

# best-fit C matching fourth derivatives at q_gamma
C_val = sp.diff(Fbeta, qv, 4).subs(qv, qg) / sp.diff(Fhalf, qv, 4).subs(qv, qg)
D4_original = sp.simplify(sp.diff(Fbeta - C_val*Fhalf, qv, 4))
probe_point = qg + sp.Rational(1, 10)
val_original = float(D4_original.subs(qv, probe_point))
print(f"D_q^4(F_beta - C*F_half) at probe point, BEFORE perturbation: {val_original}")

# perturb: agree with F_beta to order N=6 at q_gamma, arbitrary G elsewhere
G = sp.sin(5*qv) + 2
N = 6
Fbeta_tilde = Fbeta + (qv - qg)**(N+1)*G

print("\nDerivative agreement check (orders 0..N, must all be 0):")
for j in range(N+1):
    diff_j = sp.simplify(sp.diff(Fbeta_tilde - Fbeta, qv, j).subs(qv, qg))
    print(f"  order {j}: {diff_j}")

C_val_tilde = sp.diff(Fbeta_tilde, qv, 4).subs(qv, qg) / sp.diff(Fhalf, qv, 4).subs(qv, qg)
D4_tilde = sp.simplify(sp.diff(Fbeta_tilde - C_val_tilde*Fhalf, qv, 4))
val_tilde = float(D4_tilde.subs(qv, probe_point))
print(f"\nD_q^4(F_beta_tilde - C_tilde*F_half) at same point, AFTER perturbation: {val_tilde}")
print(f"Still generic/nonzero despite sharing 6 derivatives? {abs(val_tilde) > 1e-6}")
```

Output:

```
D_q^4(F_beta - C*F_half) at probe point, BEFORE perturbation: 46.93821643404021

Derivative agreement check (orders 0..N, must all be 0):
  order 0: 0
  order 1: 0
  order 2: 0
  order 3: 0
  order 4: 0
  order 5: 0
  order 6: 0

D_q^4(F_beta_tilde - C_tilde*F_half) at same point, AFTER perturbation: 49.02760824764299
Still generic/nonzero despite sharing 6 derivatives? True
```

This example uses a generic real-analytic F_β, not $\zeta(\beta + it_n(\cdot))$ itself, in accordance with the scope of Proposition 14.1 as clarified after Corollary 14.3.

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